AI Blind Assist Robot – Class 7 Working Model Project

Objective:
To build a simple AI-based assistive robot that helps a blind person walk safely, detect obstacles, warn them using voice, and assist while crossing roads.

What This Robot Can Do:

1. Detect obstacles using ultrasonic sensors
2. Give voice warnings like 'Stop', 'Obstacle ahead'
3. Assist during road crossing using IR sensors
4. Use buzzer + vibration motor for alerts
5. Voice assistant using recorded sound modules
6. Completely working hardware model
7. Easy Arduino code suitable for Class 7 students

Components Required (with Quantity):

1. Arduino UNO – 1
2. Ultrasonic Sensor (HC-SR04) – 2
3. IR Sensor Module – 2
4. DFPlayer Mini (Voice Module) – 1
5. Speaker (8 ohm) – 1
6. Buzzer – 1
7. Vibration Motor – 1
8. L298N Motor Driver – 1
9. DC Motors – 2
10. Robot Wheels – 2
11. Castor Wheel – 1
12. Rechargeable Battery (9V or 7.4V) – 1
13. Switch – 1
14. Jumper Wires – 20+
15. Breadboard – 1
16. Robot Chassis – 1
17. USB Cable – 1

How It Works (Simple Explanation):

The ultrasonic sensors continuously check the distance in front of the blind person. If an obstacle is detected within 50 cm, the robot stops and gives a voice warning. IR sensors help detect road crossing lines. When road crossing is detected, the robot gives a safe crossing message. The DFPlayer Mini plays recorded human voice messages.

Step-by-Step Building Process:

1. Fix Arduino, motor driver, and battery on the robot chassis
2. Connect motors to the motor driver
3. Attach ultrasonic sensors in front
4. Attach IR sensors at the bottom

5. Connect buzzer and vibration motor
6. Connect DFPlayer Mini and speaker
7. Upload Arduino code
8. Test obstacle detection
9. Test voice warnings
10. Test road crossing assistance

Voice Messages to Record:

1. Obstacle ahead, please stop
2. You can move forward
3. Road crossing detected, please walk carefully
4. Warning, vehicle nearby

Sample Arduino Logic (Concept):

If distance < 50cm → Stop motors → Play warning voice → Activate vibration.
If distance > 50cm → Move motors forward.
If IR detects white line → Play road crossing message.

Why This Is an AI Project:

This project uses decision-making logic similar to AI. The robot senses the environment, analyzes the situation, and responds intelligently using voice and movement.

Safety & Learning Outcome:

This project teaches students about sensors, robotics, Arduino programming, assistive technology, and real-world problem solving. It is safe, low-cost, and innovative.